



Seaborn for Data Science

Day 6 — Lecture Notes & Practical Tasks

Relational · Distribution · Categorical · Heatmaps · Pairplots

1. Introduction to Seaborn & Relational Plots

Seaborn is a statistical plotting library built on Matplotlib. It works perfectly with Pandas DataFrames and produces beautiful charts with minimal code.

1.1 Scatter and Line Plots

```
import seaborn as sns
import matplotlib.pyplot as plt

# Load sample dataset
tips = sns.load_dataset('tips')

# Scatter Plot with Hue
sns.scatterplot(data=tips, x='total_bill', y='tip', hue='time',
               style='smoker', size='size')
plt.title("Tips vs Total Bill")
plt.show()

# Line Plot with automatic confidence intervals
fmri = sns.load_dataset('fmri')
sns.lineplot(data=fmri, x='timepoint', y='signal', hue='event')
plt.title("FMRI Signal over Time")
plt.show()
```

1.2 Tasks — Relational Plots

Task 1: Load the 'penguins' dataset from Seaborn. Create a scatter plot of bill_length_mm vs bill_depth_mm, colored (hue) by species.

Task 2: Load the 'flights' dataset. Create a line plot showing the number of passengers over the years. Seaborn will automatically average the months.

2. Distribution and Categorical Plots

These plots are critical for understanding how data is spread and comparing different categories.

2.1 Histograms, KDEs, and Categorical Plots

```
import seaborn as sns
import matplotlib.pyplot as plt

penguins = sns.load_dataset('penguins')
tips = sns.load_dataset('tips')

# Distribution: Histogram + KDE
sns.histplot(data=penguins, x='body_mass_g', kde=True, hue='species')
plt.title("Body Mass Distribution")
plt.show()

# Categorical: Countplot (frequency)
sns.countplot(data=tips, x='day', hue='sex')
plt.title("Count of Customers by Day and Sex")
plt.show()

# Categorical: Boxplot (spread and outliers)
sns.boxplot(data=tips, x='day', y='total_bill', hue='smoker')
plt.title("Total Bill Distribution by Day")
plt.show()
```

2.2 Tasks — Distribution & Categorical

Task 1: Using the 'titanic' dataset, create a histogram of passenger ages. Add a KDE line and hue by 'survived'.

Task 2: Create a countplot showing the number of passengers in each 'pclass' on the Titanic.

Task 3: Create a boxplot showing the 'fare' distribution across different 'pclass' categories.

3. Matrix Plots & Pairplots

Heatmaps are perfect for viewing correlation matrices. Pairplots show relationships across all numeric variables in a dataset at once.

3.1 Heatmaps and Pairplots

```
import seaborn as sns
import matplotlib.pyplot as plt

titanic = sns.load_dataset('titanic')

# Heatmap (Correlation Matrix)
numeric_df = titanic.select_dtypes(include='number')
```

```
corr = numeric_df.corr()

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f',
            linewidths=0.5)
plt.title("Titanic Correlation Heatmap")
plt.show()

# Pairplot
penguins = sns.load_dataset('penguins')
sns.pairplot(penguins, hue='species')
plt.show()
```

3.2 Tasks — Matrix & Pairplot

Task 1: Calculate the correlation matrix for the 'tips' dataset. Plot it as a heatmap with annotations (annot=True) and a 'viridis' colormap.

Task 2: Create a pairplot for the 'iris' dataset, coloring the points by 'species'. Which species is most easily separable?

4. Comprehensive EDA Challenge

Combine Matplotlib and Seaborn for a full exploratory data analysis.

4.1 End-to-End Analysis

Load the 'diamonds' dataset from seaborn. (a) Create a countplot of diamond 'cut' types. (b) Use a boxplot to compare the 'price' across different 'color' grades. (c) Create a scatterplot of 'carat' vs 'price', hue by 'clarity'. (d) Plot a correlation heatmap for all numeric features. (e) Save your best plot as 'diamonds_analysis.png' using plt.savefig().